



ENSE 805 Project report-out & lessons learned

Project name

Project F

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Project sponsor & course facilitator

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Business need/opportunity

Children frequently encounter animals in their daily lives but often lack the knowledge and awareness to interact with them safely. Improper interactions can lead to injuries, fear-based behaviors in animals, and even increased surrender rates to shelters due to misunderstood behavior. The Regina Humane Society is committed to educating the community on responsible pet ownership and animal welfare. To support this mission, the organization is developing an interactive digital game designed to teach children the fundamentals of safe animal interactions. The game will be open source, providing schools with a structured educational resource. By incorporating real-world scenarios, children will learn essential skills such as:

- Asking permission before petting a dog.
- Recognizing animal body language.
- Avoiding behaviors that may make an animal uncomfortable or aggressive.

Inspired by the Zoe and Molly online safety game, this project will create an engaging, interactive, and scalable tool for classrooms across Canada.

The business need for this project arises from several key factors:

- Lack of structured education on animal safety in schools.
- Increase in pet ownership, leading to more human-animal interactions.
- The need for a scalable, engaging learning tool that integrates seamlessly with existing educational platforms.
- Aligning with the Regina Humane Society's mission to promote responsible pet ownership and animal welfare.

Reflections on project planning

- State and discuss the United Nation's (UN) Sustainable Development Goals (SDGs) selected and your "why" for selecting the one(s) you did

SDG 3: Good Health & Well-being

The game facilitates safer interaction between pets and children, directly contributing to the goal of reduced pet-related injury. Through child education about correct animal behavior and safety, PetCuesGame reduces the chances of accidents and fosters a healthier community. This goal is fundamental because it promotes



avoidance of injury and establishment of welfare, particularly in areas where children engage with pets frequently.

SDG 4: Quality Education

PetCuesGame is a learning app that is designed to teach children valuable life skills pertaining to animal safety. That it is interactive and gamified means it makes learning pet safety both enjoyable and effective, enhancing the quality of education by providing an engaging and accessible learning experience. SDG 4 entails ensuring inclusive and equitable quality education, and PetCuesGame does this by offering a special learning experience for children in regions with low access to proper pet safety education.

SDG 15: Life on Land

The game inculcates responsible pet ownership, which is towards a greater goal of protecting and preserving life on earth. By educating children about understanding and honoring animals' cues, PetCuesGame encourages culture of responsible animal care and maintains biodiversity. It is particularly necessary in societies with much pet-pet contact because responsible pet ownership there can have a positive influence on the local ecosystem as well as on the health of pets.

"Why" These SDGs Were chosen

These three SDGs were selected because they are most closely related to the mission of PetCuesGame, which is to educate children on how to responsibly and safely engage with pets. By addressing health, education, and animal welfare, the game not only aligns with global goals but also addresses real community issues. Through its promotion of consciousness and awareness, the game promotes the construction of a more responsible, safer community and contributes to the sustainability overall goals, as outlined by the United Nations.

- Discuss key findings from your community research and understanding/requirements gathering (Community characteristics and technology configuration inventory)
 - Discuss your professional opinion of the processes and documentation used in this course for project planning. Did they help/hinder and how?

Through our collaboration with the Regina Humane Society and interaction with community members, we found:

- Need for Accessible Pet Education: Many children interact with animals without understanding appropriate behavior, which leads to misunderstandings or harm.
- Community Tech Landscape: Most homes and schools have access to tablets, mobile devices, or computers, making a web-based solution ideal.
- Interactive Learning is Preferred: Parents and educators indicated that traditional teaching methods often fail to hold children's attention, while interactive games offer better engagement.

Professional Opinion on Project Planning Tools: The planning templates, milestone tracking, and requirement documentation helped structure the development process. Though occasionally time-consuming, these tools ensured clarity, accountability, and alignment with client needs. They encouraged a more organized and thoughtful approach to each phase of development.



- State selected north star & carryover customers. Why are these customers important to your project's golden circle (why, how, what)?

North Star and Carryover Customers

- North Star Customer: Children
- Carryover Customers: Parents, teachers, Regina humane society educators

These customers are vital to our project's Golden Circle:

- Why: To create a safer environment for pets and children by promoting responsible behavior
- How: Through fun, educational, interactive game-based learning
- What: A browser-based digital game teaching safe animal interaction

Children are our main users, but their learning and behavior are supported and reinforced by parents, teachers, and community educators.

- Summarize assumptions made and constraints uncovered, re: drafting an emerging picture

Assumptions and Constraints

Assumptions:

- Schools and humane societies are willing to adopt free web-based learning tools.
- Children are familiar with basic gaming interfaces.

Constraints:

- Limited resources for advanced animations and voiceovers
- Need to keep the platform device-agnostic and lightweight
- Time constraints related to academic deadlines and client review schedules

These influenced the scope of our MVP and the features prioritized during initial development.

- Discuss initial & the evolution of your technology stack selection, drafted prototypes, and initial Minimum Viable Products (MVPs)

Initial Technology Stack:

- Godot Game Engine (open-source, easy for 2D games)
- HTML5 Export for web compatibility
- GIMP and Inkscape for graphics
-

Evolved Stack (Confirmed):

- Godot 4.2 for scripting FSM pet behavior and animations
- WebGL/HTML5 for browser deployment
- Dialogue System, Quiz Engine, and Behavioral AI scripts customized for each scenario



- Provide images/screenshots where appropriate/needed

Workflow Diagram



Reflections on project results

- Discuss what you created. Provide key images/screenshots illustrating core functionality
 - Review your initial “Planning and initialization” video created for the first deliverable. How close did you come to realizing the solution/product you initially envisioned?
 - Summarize software design activities and findings. Ensure you discuss how you/your team either linked or envision links to design ideas back to topics discussed in class lectures

What We Created: PetCuesGame is an interactive, choice-based learning game that educates children on how to safely interact with pets in real-life scenarios. Built using the Godot 4.2 engine, it features a structured curriculum, realistic pet behavior via FSMs, dynamic animations, story-driven gameplay, and accessible HTML5 deployment. The game provides children with real-time decision-making scenarios, such as how to approach a stray dog or interpret a pet’s body language.



Key Features:

- Interactive safety scenarios (e.g., “Ask an adult before petting”)
- Dialogue system and behavioral AI for realistic pet interactions
- Quiz-based formative assessments
- Storyline missions like helping a lost dog or guiding animals safely
- Web deployment for school, home, and humane society use

Initial Vision vs. Final Product: In our initial planning video, we envisioned a game that blends fun gameplay with essential pet safety education, targeted at children. The final product came very close to that vision, with many of the planned mechanics like FSM-driven pet reactions, quizzes, and story-based learning successfully implemented. Some scope adjustments were made (e.g., fewer levels in the MVP), but the core experience remained intact.

Software Design Activities and Findings: Design efforts included brainstorming pet safety scenarios, mapping out choice paths for children, designing UI for dialogues/quizzes, and iterating FSM structures for pet reactions. These design choices were influenced by lecture topics such as user-centered design, formative assessment, and gamification strategies. Prototypes were tested and evolved using feedback from the Regina Humane Society.

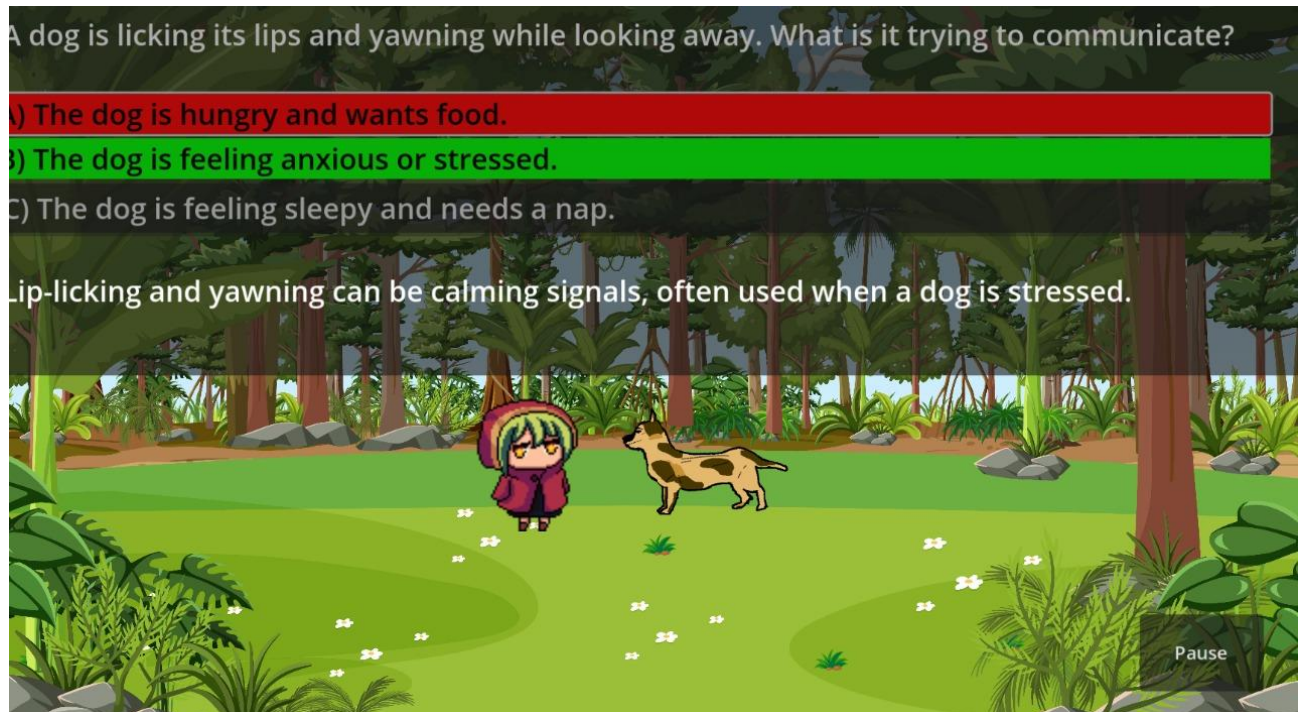
In below image you can see our first scene of the game or you can see the start screen





In some of the below screen shot you can see child and dog interaction screen





Below screen show us an final score of the game



- Summarize how you felt about this project (likes/dislikes), from your experiences with the technology stack selected, translating prototypes into real solutions, and the creation/realization of your MVPs
 - Summarize what went well during the project
 - Smooth integration of FSM and animation systems in Godot
 - High engagement and usability during test sessions with children
 - Effective collaboration with our client (Rebecca from Regina Humane Society)



- Summarize what not went well during the project
 - Time constraints limited level complexity and polish
 - Some challenges with cross-platform UI consistency
 - Scope creep early on that had to be managed
- What would you do the same on future projects?
 - Early and ongoing client feedback
 - Use of open-source tools like Godot and GIMP for cost-efficiency
 - Narrative-based design for better retention
- What would you do differently on future projects?
 - Allocate more time for user testing and polish
 - Simplify design early on to avoid over-scoping
 - Use version control systems more efficiently across the team
- Discuss opportunities and design ideas for future work
 - Add more advanced levels covering other animals (cats, rabbits, birds)
 - Include multiplayer or cooperative learning mode
 - Offer teacher dashboards with analytics and progress tracking
 - Integrate AR features to bring pets into real-world settings
 - Expand language support for accessibility in multicultural communities

General reflections on the class & project experience

- Before taking ENSE 805, were you aware of the UN SDGs?
 - No - I did not have any knowledge regarding the United Nations Sustainable Development Goals (UN SDGs) prior to attending ENSE 805. This was my first exposure to them during this course, and it really opened my eyes to how software engineering can be used to contribute meaningfully towards bringing solutions to problems globally. Through the lectures and discussions, I realized how technology and digital design could help in reaching objectives like quality education, reduced inequalities, and sustainable consumption. It helped me to understand that as a software engineer, not only can I make contributions beyond technical ones—but I can also create a more just and sustainable future.
- Typically, before taking this class, when you engineered software solutions, were you concerned with areas encompassing the UN SDGs?
 - No - Before I took this course, I wasn't exactly thinking about the UN Sustainable Development Goals (SDGs) while designing or building software. My thoughts were functional, performance-driven, and user-focused. But now that I have learned about the SDGs in ENSE 805, I realize now that some of the things I have done can actually be working towards the SDGs.



For example, I'm currently developing an educational game called PetCuesGame, which teaches children how to safely interact with animals. This game contributes to SDG 4: Quality Education by promoting early learning through digital tools, and also touches on SDG 15: Life on Land by encouraging respect and awareness for animals in the community. Although I was not making it intentionally align with the SDGs at first, now I know that how powerful the connections can be between my activities and the SDGs.

- Did learning about the UN SDG(s) help you understand better your role and responsibility as an engineer to society?
 - Yes - Learning the UN Sustainable Development Goals (SDGs) has greatly assisted me in realizing my larger responsibility as a software engineer. Previously, I was more concerned with the technical side of engineering-i.e., performance, functionality, and user experience. But after learning the SDGs, now I realize that as an engineer, my contribution can impact the problems of society-ranging from encouraging education and equality to encouraging environmental sustainability.

For instance, by developing initiatives like PetCuesGame, I can contribute to SDG 4: Quality Education by rendering educational resources interactive. Further, studying the SDGs has caused me to reflect on how I can ensure responsible choices in the software that I code so that it not only serves its consumers but the world as well. This new philosophy has motivated me to explore how I can create solutions that align with worldwide goals and serve the general good.

- What was your experience(s) in engineering your specific software solution to address the UN SDG(s) selected?

Ans:

It has been enriching and enlightening to develop PetCuesGame to address the UN SDG 4: Quality Education and SDG 15: Life on Land. I wasn't originally considering how my project can be associated with the SDGs, but as the course made me learn more about them, I realized how much my project can contribute to these global goals.

Designing the game made me ponder even more intensely about how education shapes the future. I added educational content teaching children how to safely engage with animals, cultivating respect and empathy. I further pondered creating educational materials that are accessible and fun because it encourages involvement and leads to improved learning.

The challenge was finding a way to make the game both fun and educational, while balancing game mechanisms with useful information about animal safety. It opened my eyes to the potential software engineering has in making a positive impact on the individual user, as well as the broader community. It was inspiring to find that software solutions, no matter how small, can help the world become a more sustainable and compassionate place.



- As a future engineer, what are your thoughts on the UN SDGs as a whole? Do you think they can help or hinder our work as software engineers?

Ans:

As a prospective engineer, I believe that the UN SDGs offer a strong vision for contextualizing software and technology development in a manner that supports global well-being. The goals offer a succinct set of values for building solutions to respond to some of the most significant issues confronting the world, including climate change, inequality, and education. The goals give software engineers the chance to see beyond the development of usable products and to reflect on how our work can be leveraged to facilitate beneficial change for the greater good.

In my opinion, the SDGs won't hinder our work; in fact, they can provide a sense of direction and purpose. They challenge us to think creatively and innovatively, finding solutions that not only meet user needs but also align with global challenges. For example, developing solutions that address SDG 4: Quality Education can motivate engineers to create engaging educational tools, while SDG 15: Life on Land encourages more sustainable and ethical approaches to animal-related software.

Ultimately, the SDGs can inspire engineers to be more socially responsible and to take ownership of the impact their software has on society and the environment. By integrating these goals into our work, we're not just creating products but contributing to a better world.

- Should we use the UN SDGs to guide our work or is our work dependent on customer requests, regardless of the UN SDGs?

Ans:

In my opinion, we should strive to balance both the UN SDGs and customer requests. While customer needs are paramount to any project, aligning our work with the UN SDGs can provide long-term value—not just for the customer, but for society as a whole. The SDGs offer a global framework for addressing pressing issues like inequality, climate change, and education, which software engineers can help tackle with their skills.

That being said, customer requests are still critical since they reflect the specific needs of users and businesses. The challenge lies in finding a way to integrate the SDGs with customer requirements, where possible. For example, if a customer requests an educational tool, we can ensure it aligns with **SDG 4: Quality Education**, adding features that enhance learning and engagement. Similarly, if the work involves creating products with environmental implications, we should consider **SDG 13: Climate Action** and **SDG 15: Life on Land**.

Ultimately, the SDGs don't have to be separate from customer requests; they can enhance and enrich the solutions we build, creating software that serves both the client's needs and broader societal goals.

- Will you use your understanding of the UN SDGs in engineering solutions in the future?
 - Yes - For sure, I plan to use my understanding of the UN SDGs in engineering solutions in the future. Learning about these global goals has opened my eyes to the role of software engineers in society. Now I know that our work can potentially cause not just technical change but also social change. Moving ahead, I will actively seek opportunities to align the SDGs with my work, be it creating solutions for SDG 4: Quality Education, mitigating SDG 13: Climate Action, or supporting SDG 15: Life on Land. These goals



offer a powerful framework to guide the impact of technology in a positive manner, and I believe software engineers have a unique opportunity to contribute meaningfully to these global goals.

- Will your experience learning about the UN SDGs inform your career path decisions in the future?
 - Yes - Learning about the UN SDGs has indeed impacted my understanding of my possible future profession as an engineer. While before, technical competency and goals that have to do with the project per se filled my mind, I now realize that there is a broader responsibility in being an engineer—it's one to create solutions for the overall good of society. In the future, I will pay greater attention to how my work feeds into global objectives like SDG 4: Quality Education, SDG 13: Climate Action, and SDG 15: Life on Land. The new perspective has motivated me to look for projects not just that address client needs but also that support sustainable development and moral behavior. I now have an obligation to choose career options and projects that allow me to use technology for social good.
- Provide any other comments on the project

PetCuesGame has strong potential to evolve into a widely-used educational tool in schools, shelters, and public campaigns. By reinforcing empathy and responsible pet ownership from an early age, the game contributes meaningfully to both community safety and animal welfare.